

Econometrics

Time Series Analysis

This website is built around the slides from my lectures, with the full course narrative woven around them so that the site also works as a self-contained reference for the module. Throughout, you'll find signposts to other textbooks and animations, along with R graphs, audio snippets, and other resources where relevant.

If there's any concept you'd like more support with, just let me know. The syllabus is fixed, but I'm always happy to add supplementary materials like an audio snippet or a screen recording.

Topic 01

Introducing Time Series Analysis

Core statistical foundations, what time series analysis is, and why it needs special techniques.

[Read notes](#) → [PDF](#) ↓

Topic 02

Introducing ARMA(p, q) models

White noise, MA and AR building blocks, the lag operator, and the full ARMA(p, q) family.

[Read notes](#) → [PDF](#) ↓

Topic 03

Measuring dependence

Autocovariance and autocorrelation functions — how we quantify dependence across time.

[Read notes](#) → [PDF](#) ↓

Topic 04

Defining stationarity

Strong and weak stationarity, and the autocovariance structure of stationary series.

[Read notes](#) → [PDF](#) ↓

Coming soon

More topics

Further topics will be added here as the module progresses.